

4. Force System Resultants

Tuesday, September 3, 2024 9:39 AM

* Shift from a particle analysis to a rigid body analysis (bodies that do not deform under loading)

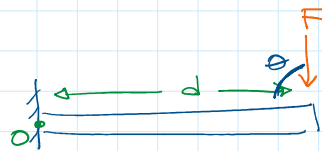
* Leverage

* Application of force matters (position of force on body)

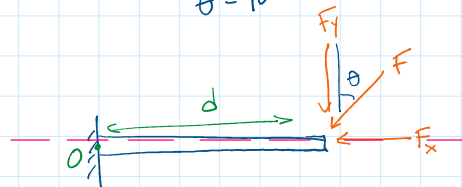
* Bending Moment

$$M_o = F \cdot d \text{ [N}\cdot\text{m]}$$

Perpendicular Vectors
(Force & Position)



$$\theta = 90^\circ$$



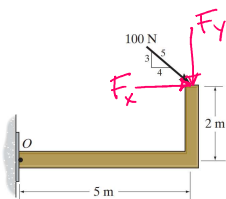
M_o due to F_x is zero because F_x 's line of action goes through point O.

$$M_o \neq F \cdot d$$

$$M_o = F_y \cdot d$$



F4-1. Determine the moment of the force about point O.

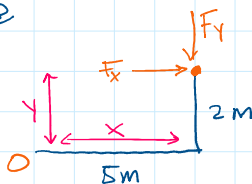


Prob. F4-1

$$M_o = F \cdot d$$

$$F_x = 80 \text{ N} \quad F_y = 60 \text{ N}$$

FBD



* perpendicular distance to F_x is y (from O)

* perpendicular distance to F_y is x (from O)

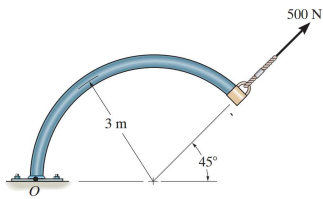
$$M_o = -F_x \cdot y - F_y \cdot x$$

$$M_o = (-80 \text{ N})(2 \text{ m}) - (60 \text{ N})(5 \text{ m})$$

$$-160 \text{ N}\cdot\text{m} - 300 \text{ N}\cdot\text{m}$$

$$M_o = 460 \text{ N}\cdot\text{m} \text{ cw}$$

F4-6. Determine the moment of the force about point O.



Prob. F4-6

$$\cos(45^\circ) = \frac{a}{3} \Rightarrow a = 3 \cos(45^\circ)$$

$$\sin(45^\circ) = \frac{y}{3} \Rightarrow y = 3 \sin(45^\circ)$$

$$x = 3 + 3 \cos(45^\circ)$$

$$y = 3 \sin(45^\circ)$$

$$\cos(45^\circ) = \frac{F_x}{F}$$

$$F_x = F \cos(45^\circ)$$

$$F_y = F \sin(45^\circ)$$

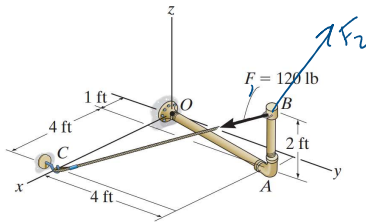
$$M_O = \underbrace{(3 + 3 \cos(45^\circ))}_{x} F_y - \underbrace{3 \sin(45^\circ)}_{y} F_x$$

$$M_O = (3 + 3 \cos 45^\circ) F \sin(45^\circ) - 3 \sin(45^\circ) F \cos(45^\circ)$$

$$M_O = (3 + 3 \cos 45^\circ) 500 \sin(45^\circ) - 3 \sin(45^\circ) (500) \cos(45^\circ)$$

$$M_O = 1060.67 \text{ N}\cdot\text{m} \quad \text{ccw}$$

F4-11. Determine the moment of force $\mathbf{F}$ about point O. Express the result as a Cartesian vector.



Prob. F4-11

$$\vec{M}_O = \vec{r}_{OB} \times \vec{F}$$

* cross product

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\vec{r}_{OB} = 1 \hat{i} + 4 \hat{j} + 2 \hat{k}$$

$$\vec{F} = 120 \hat{u}_{BC}$$

$$\vec{r}_{OC} = 5 \hat{i}$$

$$\vec{r}_{BC} = 5 \hat{i} - [1 \hat{i} + 4 \hat{j} + 2 \hat{k}]$$

$$\vec{r}_{BC} = 4 \hat{i} - 4 \hat{j} - 2 \hat{k} - [1 \hat{i}]$$

$$r_{BC} = \sqrt{4^2 + (-4)^2 + (-2)^2} = \sqrt{36} = 6 \text{ [ft]}$$

$$\hat{u}_{BC} = \frac{4}{6} \hat{i} - \frac{4}{6} \hat{j} - \frac{2}{6} \hat{k}$$

$$\hat{u}_{BC} = \frac{2}{3} \hat{i} - \frac{2}{3} \hat{j} - \frac{1}{3} \hat{k}$$

$$\vec{F} = F \cdot \hat{u}_{BC} = 120 \left(\frac{2}{3} \hat{i} - \frac{2}{3} \hat{j} - \frac{1}{3} \hat{k} \right) - [16]$$

$$\vec{F} = \underbrace{80 \hat{i}}_{F_x} - \underbrace{80 \hat{j}}_{F_y} - \underbrace{40 \hat{k}}_{F_z} - [16]$$

$$\vec{M}_O = \vec{r} \times \vec{F}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$+ (a_y b_z - a_z b_y) \hat{i} - (a_x b_z - a_z b_x) \hat{j}$$

$$-(a_x b_z - a_z b_x) \hat{j}$$

$$(a_x b_y - a_y b_x) \hat{k}$$

$$\vec{M}_O = \vec{r}_{OB} \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 80 & -80 & -40 \end{vmatrix}$$

$$\vec{M}_O = (4(-40) - 2(-80)) \hat{i}$$

$$-160 + 160$$

$$- [1(-40) - 2(80)] \hat{j}$$

$$-40 - 160$$

$$+ [1(-80) - 4(80)] \hat{k}$$

$$-80 - 320$$

$$\vec{M}_O = 200 \hat{j} - 400 \hat{k} \text{ lb}\cdot\text{ft}$$

$$(\vec{M}_O)_1 = \vec{r} \times \vec{F}_1$$

$$(\vec{M}_O)_2 = \vec{r} \times \vec{F}_2$$

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2$$

$$\vec{M}_O = (\vec{M}_O)_1 + (\vec{M}_O)_2$$

$$\vec{M}_O = \vec{r} \times \vec{F}_R$$

F4-12. If the two forces $\mathbf{F}_1 = \{100\mathbf{i} - 120\mathbf{j} + 75\mathbf{k}\}$ lb and $\mathbf{F}_2 = \{-200\mathbf{i} + 250\mathbf{j} + 100\mathbf{k}\}$ lb act at A, determine the resultant moment produced by these forces about point O. Express the result as a Cartesian vector.

$$\vec{F}_1 = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\vec{F}_1 = 100\hat{i} - 120\hat{j} + 75\hat{k}$$

$$\vec{F}_2 = -200\hat{i} + 250\hat{j} + 100\hat{k}$$

$$\vec{F}_R = -100\hat{i} + 130\hat{j} + 175\hat{k}$$

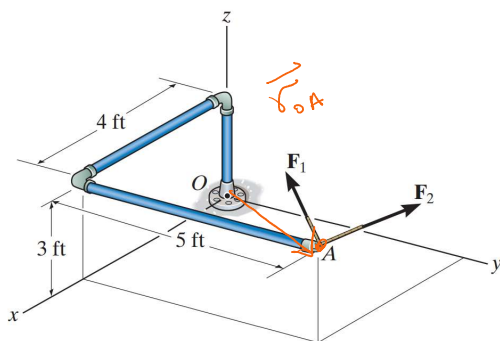
$$\vec{r}_{OA} = 4\hat{i} + 5\hat{j} + 3\hat{k}$$

$$\vec{M}_O = \vec{r}_{OA} \times \vec{F}_R$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 5 & 3 \\ -100 & 130 & 175 \end{vmatrix}$$

$$[(5)(175) - 3(130)] \hat{i}$$

$$- [4(175) - 3(-100)] \hat{j}$$

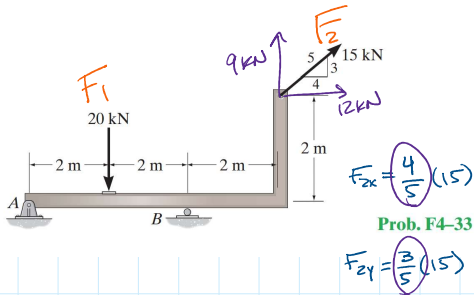


Prob. F4-12

$$[4(130) - 5(-100)] \hat{k}$$

$$\vec{M}_0 = 485\hat{i} - 1000\hat{j} + 1020\hat{k} \quad [16 \cdot ft]$$

F4-33. Replace the loading system by an equivalent resultant force and specify where the resultant's line of action intersects the horizontal segment of the member measured from A.

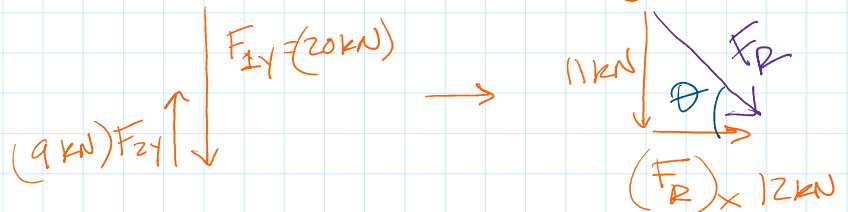


$$F_{1x} = 0$$

$$F_{1y} = 20 \text{ kN}$$

$$F_{2x} = 12 \text{ kN} = \frac{4}{5}(15 \text{ kN})$$

$$F_{2y} = 9 \text{ kN} = \frac{3}{5}(15 \text{ kN})$$



$$F_R^2 = (F_R)_x^2 + (F_R)_y^2$$

$$F_R = \sqrt{11^2 + 12^2} = \sqrt{121 + 144} = \sqrt{265}$$

$$F_R = 16.28 \text{ kN}$$

$$\theta = \tan^{-1}\left(\frac{11}{12}\right) \quad \theta = 42.5^\circ$$

* How far should F_R be from point A?

$\sum M_A = (M_A)_R \rightarrow$ The sum of the moments about A due to F_1 & F_2 should equal the moment about A due to F_R .

$$(M_A)_1 = -x_1 F_{1y} = -(2\text{m})(20\text{kN}) = -40 \text{ kN}\cdot\text{m}$$

$$(M_A)_2 = x_2 F_{2y} - y_2 F_{2x}$$

$$= (6\text{m})(9\text{kN}) - (2\text{m})(12\text{kN}) = 54 \text{ kN}\cdot\text{m} - 24 \text{ kN}\cdot\text{m}$$

$$= 30 \text{ kN}\cdot\text{m}$$

$$+\circlearrowleft \sum M_A = -40 \text{ kN}\cdot\text{m} + 30 \text{ kN}\cdot\text{m} = -10 \text{ kN}\cdot\text{m}$$

$$\sum M_A = 10 \text{ kN}\cdot\text{m} \quad \text{CW}$$

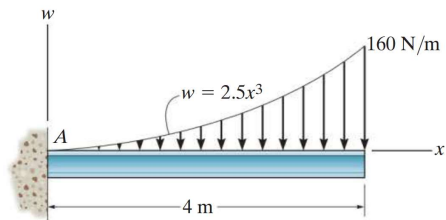
$$\sum M_A = (M_A)_R = x(F_R)_y$$

only account for component that is vertical & perpendicular to x.

$$\frac{10 \text{ kN}\cdot\text{m}}{11 \text{ kN}} = x \frac{(11 \text{ kN})}{11 \text{ kN}}$$

$$x = 0.91 \text{ m}$$

F4-42. Determine the resultant force and specify where it acts on the beam measured from A.



Prob. F4-42

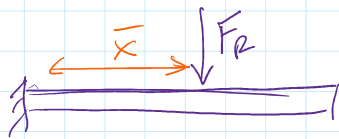
$$F_R = \int_L w(x) dx$$

$$\bar{x} = \frac{\int_L x \cdot w(x) dx}{F_R}$$

equilibrium analysis

$$F_R = \int_0^4 2.5x^3 dx = \left. \frac{5}{8}x^4 \right|_0^4 = \frac{5}{8}(4)^4 - \frac{5}{8}(0)^4$$

$$F_R = 160 \text{ N}$$



$$\bar{x} = \frac{\int_L x \cdot w dx}{F_R}$$

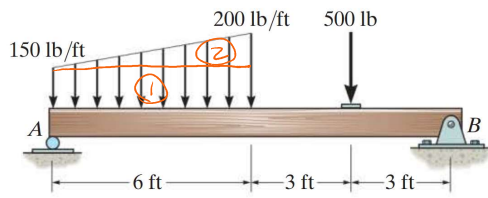
$$\bar{x} = \frac{\int_0^4 x(2.5x^3) dx}{F_R} = \frac{\int_0^4 2.5x^4 dx}{F_R}$$

$$\bar{x} = \frac{\left. \frac{1}{2}x^5 \right|_0^4}{F_R} = \frac{\frac{1}{2}(4)^5 - 0}{160}$$

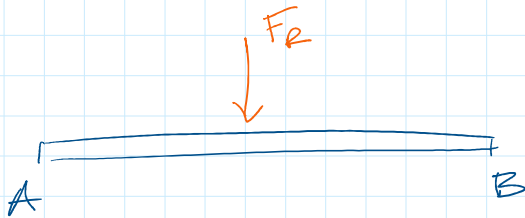
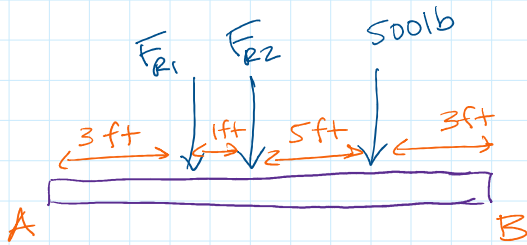
$$\bar{x} = 3.2 \text{ m}$$

F4-40. Determine the resultant force and specify where it acts on the beam measured from A.

& Get the reactions @ the supports.



Prob. F4-40



$$F_{R1} = 150 \frac{\text{lb}}{\text{ft}} (6 \text{ ft})$$

$$F_{R1} = 900 \text{ lb}$$

$$F_{R2} = \frac{1}{2} (6 \text{ ft}) (50 \frac{\text{lb}}{\text{ft}})$$

$$F_{R2} = 150 \text{ lb}$$

$$F_R = F_{R1} + F_{R2} + 500$$